



Exp-2

DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

AIM:

To add the constraints like primary key, foreign key, unique key and check using DDL commands.

Description:

PRIMARY KEY:

The PRIMARY KEY constraint uniquely identifies each record in a database table. Primary keys must contain UNIQUE values, and cannot contain NULL values.

A table can have only one primary key, which may consist of single or multiple fields.

FOREIGN KEY:

- A FOREIGN KEY is a key used to link two tables together.
- A FOREIGN KEY is a field (or collection of fields) in one table that refers to the PRIMARY KEY in another table.
- The table containing the foreign key is called the child table, and the table containing the candidate key is called the referenced or parent table.

UNIQUE Constraint:

- The UNIQUE constraint ensures that all values in a column are different.
- Both the UNIQUE and PRIMARY KEY constraints provide a guarantee for uniqueness for a column or set of columns.
- A PRIMARY KEY constraint automatically has a UNIQUE constraint.
- However, you can have many UNIQUE constraints per table, but

only one PRIMARY KEY constraint per table.

CHECK Constraint:

- The CHECK constraint is used to limit the value range that can be placed in a column
- If you define a CHECK constraint on a single column it allows only certain values for this column.
- If you define a CHECK constraint on a table it can limit the values in certain columns based on values in other columns in the row.

Questions:

- 1) Alter the table PATIENT with following structure.

#	Column Name	Constraint
1	Patient_ID	PRIMARY KEY

- 2) Alter the table name DOCTOR with following structure.

#	Column Name	Constraint
1	Doctor_ID	PRIMARY KEY
2	Gender	CHECK ('M' or 'F')

- 3) Alter the table name DEPARTMENT with following structure.

#	Column Name	Constraints
1	DeptNo	PRIMARY KEY

- 4) Alter the table name APPOINTMENT with following structure.

#	Column Name	Constraint
1	App_ID	PRIMARY KEY
2	Room_No	CHECK (Room number should not be empty)

- 5) Alter the table name PATIENT with following structure.

#	Column Name	Constraints
1	Patient_Name	UNIQUE KEY

- 6) After the PATIENT & APPOINTMENT tables are successfully created, test if you can add a constraint FOREIGN KEY to the APPOINTMENT of this table.

OUTPUTS:

1)

```
ALTER TABLE Patient
ADD PRIMARY KEY (Patient_ID);
DESC Patient;
```

Field	Type	Null	Key	Default	Extra
Patient_ID	char(5)	NO	PRI	NULL	
Patient_Name	varchar(20)	YES		NULL	
Age	int	YES		NULL	
Gender	char(1)	YES		NULL	
Disease	varchar(20)	YES		NULL	
Phone_No	bigint	YES		NULL	

2)

```
ALTER TABLE Doctor
ADD PRIMARY KEY (Doctor_ID);

ALTER TABLE Doctor
ADD CHECK (Gender IN ('M', 'F'));

DESC Doctor;
```

Field	Type	Null	Key	Default	Extra
Doctor_ID	char(5)	NO	PRI	NULL	
Doctor_Name	varchar(20)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
Specialization	varchar(20)	YES		NULL	
Phone_No	bigint	YES		NULL	
Department	varchar(20)	YES		NULL	

3)

```
ALTER TABLE Department
ADD PRIMARY KEY (Dept_ID);
```

```
DESC Department;
```

Field	Type	Null	Key	Default	Extra
Dept_ID	char(5)	NO	PRI	NULL	
Dept_Name	varchar(20)	YES		NULL	
Dept_Head	varchar(20)	YES		NULL	

4)

```
ALTER TABLE Appointment
ADD PRIMARY KEY (App_ID);
```

```
ALTER TABLE Appointment
ADD CHECK (Room_No <> '');
```

```
DESC Appointment;
```

Field	Type	Null	Key	Default	Extra
App_ID	char(5)	NO	PRI	NULL	
Patient_ID	char(5)	YES		NULL	
Doctor_ID	char(5)	YES		NULL	
App_Date	date	YES		NULL	
App_Time	time	YES		NULL	
Room_No	char(5)	YES		NULL	

5)

```
ALTER TABLE Patient
ADD UNIQUE (Patient_Name);
```

```
DESC Patient;
```

Field	Type	Null	Key	Default	Extra
Patient_ID	char(5)	NO	PRI	NULL	
Patient_Name	varchar(20)	YES	UNI	NULL	
Age	int	YES		NULL	
Gender	char(1)	YES		NULL	
Disease	varchar(20)	YES		NULL	
Phone_No	bigint	YES		NULL	

6)

```

ALTER TABLE Appointment
ADD CONSTRAINT FK_Patient
FOREIGN KEY (Patient_ID)
REFERENCES Patient(Patient_ID);

DESC Appointment;
```

Field	Type	Null	Key	Default	Extra
App_ID	char(5)	NO	PRI	NULL	
Patient_ID	char(5)	YES	MUL	NULL	
Doctor_ID	char(5)	YES		NULL	
App_Date	date	YES		NULL	
App_Time	time	YES		NULL	
Room_No	char(5)	YES		NULL	

Result:

DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.

